

Ex.No. 11

Implementation of IoT-based Smart Agriculture using MATLAB

Objective 1:

To monitor the Soil Moisture (%) level in agricultural fields using IoT-based sensing simulation in MATLAB.

CODE:

```
clc; clear; close all;

% Simulated soil moisture data (%)

time = 1:10;

soil_moisture = [30 35 28 25 20 18 22 30 40 45];

% Threshold for irrigation

threshold = 40;

figure

% ----- Graph 1: Soil Moisture Variation -----

subplot(2,1,1)

plot(time, soil_moisture, '-o', 'LineWidth', 2)

title('Soil Moisture Monitoring in IoT Smart Agriculture')

xlabel('Time (Hours)')

ylabel('Soil Moisture (%)')

grid on

hold on

yline(threshold, 'r--', 'Threshold')

% ----- Graph 2: Moisture Status (Dry/Wet Level) -----

status = soil_moisture >= threshold;

subplot(2,1,2)

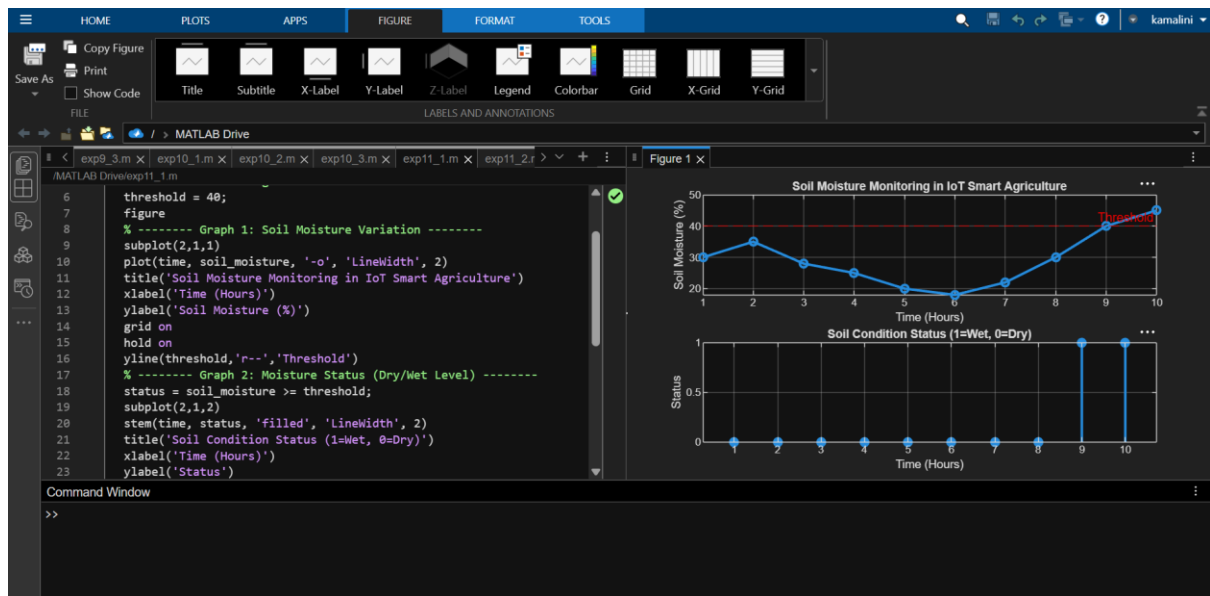
stem(time, status, 'filled', 'LineWidth', 2)

title('Soil Condition Status (1=Wet, 0=Dry)')

xlabel('Time (Hours)')

ylabel('Status')
```

grid on



Objective 2:

To control and display the Water Pump Status (ON/OFF) based on threshold soil moisture conditions.

CODE:

```
clc; clear; close all;
```

```
% Simulated soil moisture (%)
```

```
time = 1:10;
```

```
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
```

```
% Threshold level
```

```
threshold = 40;
```

```
% Pump control logic (1 = ON, 0 = OFF)
```

```
pump_status = soil_moisture < threshold;
```

```
figure
```

```
% ----- Graph 1: Soil Moisture vs Pump Status -----
```

```
subplot(2,1,1)
```

```
plot(time, soil_moisture, '-o', 'LineWidth', 2)
```

```
title('Soil Moisture vs Pump Control')
```

```
xlabel('Time (Hours)')
```

```
ylabel('Soil Moisture (%)')
```

```
grid on
```

```
hold on
```

```
ylime(threshold,'r--','Threshold')
```

```
% ----- Graph 2: Pump Status (ON/OFF) -----
```

```
subplot(2,1,2)
```

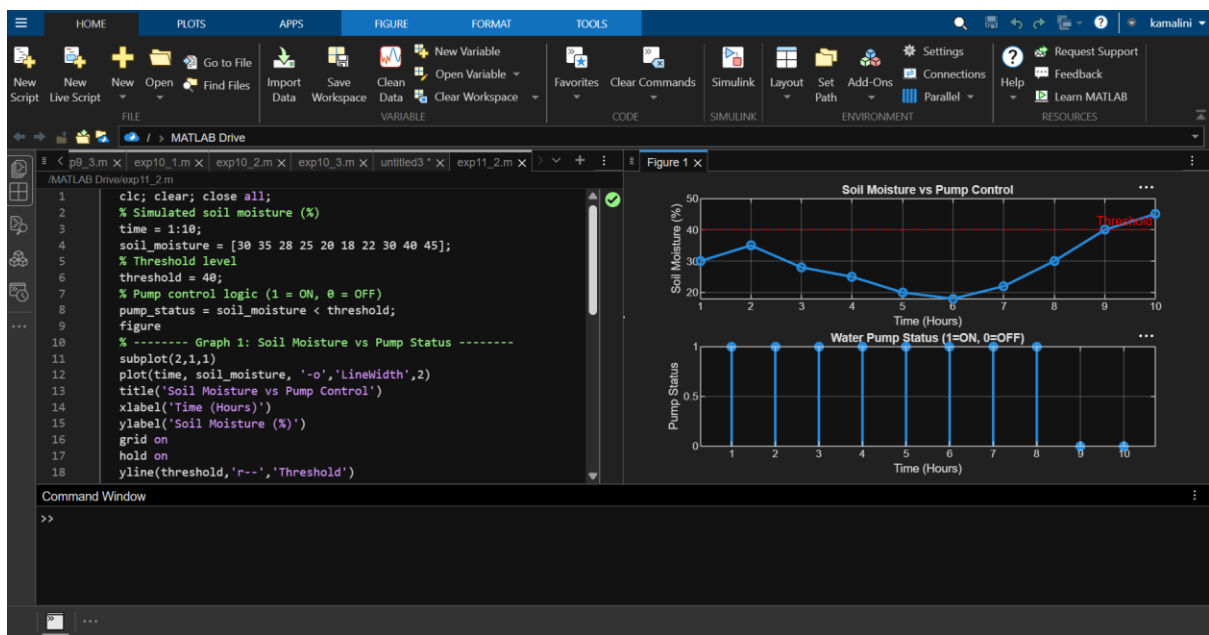
```
stem(time, pump_status, 'filled','LineWidth',2)
```

```
title('Water Pump Status (1=ON, 0=OFF)')
```

```
xlabel('Time (Hours)')
```

```
ylabel('Pump Status')
```

```
grid on
```



Objective 3:

To evaluate the Irrigation Efficiency (%) by analyzing water usage effectiveness and moisture maintenance level.

CODE:

```
clc; clear; close all;
```

```
% Simulated soil moisture data (%)
```

```
time = 1:10;
```

```

soil_moisture = [30 35 28 25 20 18 22 30 40 45];

% Threshold level

threshold = 40;

% Irrigation efficiency calculation (%)

efficiency = (soil_moisture ./ threshold) * 100;

efficiency(efficiency > 100) = 100; % limit to

100%

```

```

figure

% ----- Graph 1: Soil Moisture vs Efficiency

----- subplot(2,1,1)

plot(time, soil_moisture,

'-o','LineWidth',2) title('Soil Moisture

Variation')

xlabel('Time (Hours)')

ylabel('Soil Moisture (%)')

grid on

hold on

yline(threshold,'r--','Threshold')

% ----- Graph 2: Irrigation Efficiency

----- subplot(2,1,2)

plot(time, efficiency, '-s','LineWidth',2)

title('Irrigation Efficiency (%) in Smart

Agriculture') xlabel('Time (Hours)')

ylabel('Efficiency (%)')

grid on

```

